

W5500 Module — Measurement Request

Written in ASD-STE100 Simplified Technical English.

1. Why we need this data

We build a printed circuit board. The W5500 module attaches to this board.

The module has two rows of pins. We must know the distance between the two rows. We must know this distance before we make the board.

The manufacturer does not give this distance. The data sheet does not have it.

IMPORTANT

Do not send a photograph of the module with a ruler. We tried this two times. It did not work.

A photograph gives an error of 2 mm to 7 mm. The distance that we must find changes in steps of 2.54 mm. Thus the error is larger than the step. A photograph cannot give the correct answer.

2. Procedure A — count the holes

This procedure gives an exact answer. You do not measure a length. You count holes.

1. Get a breadboard. You can also use a piece of perfboard. The holes must have a pitch of 0.1 inch (2.54 mm).
2. Push the pins of the module into the holes. Push the module down fully.
3. Select one pin in the first row.
4. Find the pin in the second row that is directly opposite this pin.
5. Count the empty holes between the two pins.
6. Record this number.

Send us the number only. Do not measure a distance.

3. Procedure B — find the direction of the socket

1. Hold the module. The pins must point down.
2. Turn the module until the Ethernet socket points at you.
3. Look at the two rows of pins.
4. Tell us the direction of the rows. The rows go from left to right, or the rows go away from you.

4. Procedure C — identify the module

1. Read the text that is printed on the circuit board.
2. Send us this text. You can also send a photograph of the text.

The module is one of two types. It is a WIZnet WIZ850io. It is also possible that it is a W5500 Lite copy or a USR-ES1 copy.

The two types have the same pin functions. The two types do not have the same shape. Thus we must know the type.

5. What we do with your data

Data from procedure A

The pitch of the pins is 2.54 mm. This value is a standard. Thus we calculate the distance:

$$\text{distance} = (\text{number of holes} + 1) \times 2.54 \text{ mm}$$

HOLES BETWEEN THE ROWS	DISTANCE	NOTE
0	2.54 mm	The two rows touch. This is one 2 × 6 connector.
5	15.24 mm	
6	17.78 mm	
7	20.32 mm	This is the value that our design uses now.
8	22.86 mm	This is the same as the ESP32 board.

Data from procedure B

We turn the module on the circuit board. The Ethernet socket must point away from the board. Your data tells us the angle.

Data from procedure C

Your data tells us if the distance is correct for our module.

6. If you do not have a breadboard

1. Get a caliper.
2. Measure from the center of one pin to the center of the pin opposite it.
3. Send us the value. Give one number after the decimal point.

We round your value to the nearest multiple of 2.54 mm. The correct value is always a multiple of 2.54 mm.

After we get your answer, we change one line in the file `gen_pcb.py`. Then we make the board again and we test it. This operation takes 10 seconds. We do not order the boards before we have your answer.